MIL-F-275C
13 December 1982
SUPERSEDING
MIL-I-275B
6 May 1969

MILITARY SPECIFICATION

FERRIC OXIDE, BLACK, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four classes of technical grade black ferric oxide (Fe_3O_4).

1.2 Classification. Ferric oxide shall be of the following classes as specified (see 6.2):

Class A - 80 mesh
Class B - 12 mesh
Class C - 100 mesh
Class D - 200 mesh

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve Test

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D1193 - Reagent Water
D3872 - Ferrous Iron in Iron Oxides

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical characteristics. Ferric oxide shall conform to the chemical characteristics of table I when tested as specified therein.

3.2 Granulation characteristics. Ferric oxide shall conform to the applicable granulation characteristics of table II when tested as specified in 4.2.4.5.

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TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Moisture	----	0.20	4.2.4.1
Metallic iron	----	1.00	4.2.4.2
Ferrous oxide (FeO)	----	22.0	4.2.4.3
Total iron	69.0	----	4.2.4.4

TABLE II. Granulation characteristics

Sieve size	Percent by weight passing			
	Class A	Class B	Class C	Class D
No. 12, minimum	----	99.9	----	----
No. 20, minimum	----	95	----	----
No. 50, maximum	----	70	----	----
No. 80, minimum	99.0	----	----	----
No. 100, minimum	----	----	99.0	----
No. 140, maximum	----	10	----	----
No. 200, minimum	----	----	----	99.0
No. 270, maximum	----	----	50	----

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the ferric oxide of one class, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling shall be conducted in accordance with table III. A representative specimen of approximately 1 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container damaged or leaking
(c) Unitization not as specified
(d) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Moisture. Heat a moisture dish in an oven at 105° to 110°C for 1 hour, cool in a desiccator, and weigh to the nearest 0.1 milligram (mg). Transfer approximately 10 grams (g) of the specimen to the dish, stopper, and weigh to the nearest 0.1 milligram. Heat unstoppered in an oven at 105° to 110°C for 1 hour, cool to room temperature in a desiccator, replace the stopper, and weigh to the nearest 0.1 mg. Repeat the heating for 1/2-hour periods until successive weighings differ by no more than 0.1 mg, but do not continue heating for more than 3 hours. Calculate the loss in weight as percent by weight moisture as follows:

$$\text{Percent by weight moisture} = \frac{100\ (A - B)}{(A - C)}$$

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where: A = Weight of dish and specimen before heating in grams,
B = Weight of dish and specimen after drying to constant weight in grams, and
C = Weight of empty, dry dish in grams.

4.2.4.2 Metallic iron.

(a) Zimmerman-Reinhardt reagent. Dissolve 51 g of manganous sulfate, monohydrate ($MnSO_4 \cdot H_2O$) in 300 milliliters (ml) of water. Add a cooled solution of 130 ml of concentrated sulfuric acid in 300 ml of water and 138 ml of concentrated phosphoric acid. Dilute to 1 liter with water.

(b) Procedure. Grind 3 to 4 g of the specimen to pass a No. 40 sieve. Weigh to the nearest 0.1 mg approximately 2 g of the ground specimen into a 250-ml Erlenmeyer flask. Add 3 g of mercuric chloride, displace the air from the flask with a continuous flow of carbon dioxide and add 100 ml of boiling water. Place on a steam bath for 10 minutes and agitate frequently while continuing the flow of carbon dioxide. Remove from the steam bath, cool to room temperature, and discontinue the flow of carbon dioxide. Filter, and wash the residue with water, catching the filtrate and washings in a beaker containing 25 ml of Zimmerman-Reinhardt reagent prepared as specified in (a) and 350 ml of water. Titrate immediately with 0.1N potassium permanganate solution. Calculate the percent by weight metallic iron as follows:

$$\text{Percent by weight metallic iron} = \frac{5.585\ AB}{W}$$

where: A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution, and
W = Weight of specimen in grams.

4.2.4.3 Ferrous oxide. Determine the ferrous oxide content of the specimen in accordance with ASTM D3872.

4.2.4.4 Total iron. Grind approximately 1 g of the specimen to pass a No. 40 sieve. Weigh to the nearest 0.1 mg approximately 0.2 g of the ground specimen into a porcelain crucible weighed to the nearest 0.1 mg. Ignite at dull red heat for at least 1 hour. Cool to room temperature, add 5 ml of concentrated hydrochloric acid, cover with a watch glass, and place on a steam bath. As the characteristic yellow color of ferric chloride forms, decolorize by adding dropwise, and with stirring, a solution made by dissolving 50 g of stannous chloride, dihydrate ($SnCl_2 \cdot 2H_2O$) in 100 ml of concentrated hydrochloric acid and diluting to 1 liter with water. (Note: A large excess of stannous chloride leads to the formation of black compounds which spoil the analysis, or to excessive amounts of mercurous chloride which leads to high results. If the proper amount of stannous chloride has been added, a slightly silky precipitate of mercurous chloride is obtained.) Continue the dropwise addition of the stannous chloride solution as the ferric chloride color forms until all of the specimen is dissolved and the solution is completely decolorized. Add one or

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two drops of the stannous chloride solution in excess. Remove from the steam bath, cool to room temperature, and add at one stroke, 15 ml of 10-percent mercuric chloride solution. Allow to stand for 2 minutes covered with a watch glass. Wash the contents of the crucible into a beaker containing 25 ml of Zimmerman-Reinhardt reagent prepared as specified in 4.2.4.2(a) and 400 ml of water. Titrate immediately with 0.1N potassium permanganate solution. Calculate the percent by weight total iron as follows:

$$\text{Percent by weight total iron} = \frac{5.585AB}{W}$$

where: A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution, and
W = Weight of specimen in grams.

4.2.4.5 Granulation characteristics. Select sieves conforming to RR-S-366 as specified in table II. Weigh the dry sieves and a bottom pan. If more than one sieve is specified, nest the sieves in order of increasing fineness with the coarsest sieve on top. Place the nest of sieves on the bottom pan. Weigh to the nearest 0.01 g approximately 100 g of the specimen and transfer to the top sieve. Cover and shake for 3 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the specimen retained on each sieve and on the bottom pan and calculate the percent specimen passing through each sieve.

5. PACKAGING

5.1 Unit packing, industrial. A quantity of 50 (+1/2 or -0) pounds of ferric oxide shall be unit packed in accordance with MIL-STD-1188.

5.2 Packing. Ferric oxide, unit packed as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Unitization. Ferric oxide, packed as specified above, shall be unitized in a manner to assure acceptance by the common carrier applicable to the mode of transportation, safe delivery to destination, and stable stacking in storage.

5.4 Marking. Containers shall be marked in accordance with MIL-STD-1188. Unitized loads shall be marked to assure arrival at destination.

6. NOTES

6.1 Intended use. Ferric oxide is intended for the uses listed in table IV.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification and the class of ferric oxide required (see 1.2).

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TABLE IV. Intended uses

Classification	For use in the manufacture of
Class A	Pyrotechnic mixtures such as CN candle starter mixture
Class B	Thermite
Class C	First fire mixtures
Class D	Starter mixtures

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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